

Course Material

Discrete Mathematics and Graph Theory

(BMAT205L)

Ankush Chanda

ACKNOWLEDGEMENT

CONTENTS

1	Counting Techniques	2
1.1	Recurrence Relation	2
1.2	Generating Functions	2
1.3	Exercises	5

1 COUNTING TECHNIQUES

1.1 RECURRENCE RELATION

Definition 1.1. (Recurrence Relation) An equation that expresses the general term of the sequence $\{a_n\}$ in terms of one or more of the previous terms of the sequence, namely $a_0, a_1, \dots, a_{n-1}$, for all integers n with $n \geq n_0$, where n_0 is a non-negative integer, is called a recurrence relation for $\{a_n\}$ or a difference equation.

If the terms of a sequence satisfy a recurrence relation, then the sequence is called a solution of the recurrence relation.

Definition 1.2. A recurrence relation of the form $c_0 a_n + c_1 a_{n-1} + c_2 a_{n-2} + \dots + c_k a_{n-k} = f(n)$ is called a linear recurrence relation of degree k with constant coefficients, where $c_0, c_1, \dots, c_k$ are real numbers and $c_k \neq 0$. The recurrence relation is called linear, because each a_r is raised to the power 1 and there are no products such as $a_r \cdot a_s$. Since a_n is expressed in terms of the previous k terms of the sequence, the degree or order of the recurrence relation is said to be k .

In other words the degree is the difference between the greatest and least subscripts of the members of the sequence occurring in the recurrence relation. If $f(n) = 0$, the recurrence relation is said to be homogeneous; otherwise it is said to be non-homogeneous.

Example 1.1. a. We consider the famous Fibonacci sequence $0, 1, 1, 2, 3, 5, 8, 13, \dots$ which can be represented by the recurrence relation

$$F_{n+2} = F_{n+1} + F_n,$$

where $n \geq 0$ and $F_0 = 0, F_1 = 1$. This recurrence relation is a linear homogeneous recurrence relation with constant coefficients and of degrees 2.

b. For the geometric progression $4, 12, 36, 108, \dots$, the common ratio is 3. If $\{a_n\}$ represents this infinite sequence, we see that $\frac{a_{n+1}}{a_n} = 3$ which implies $a_{n+1} = 3a_n, n \geq 0$ is the recurrence relation corresponding to the geometric sequence $\{a_n\}$.

1.2 GENERATING FUNCTIONS

Definition 1.3. (Generating Functions) The generating function of a sequence $a_0, a_1, a_2, \dots$ is the expression

$$G(x) = a_0 + a_1 x + a_2 x^2 + \dots = \sum_{n=0}^{\infty} a_n x^n.$$

Example 1.2. a. The generating function for the sequence $1, 1, 1, 1, \dots$ is given by

$$G(x) = 1 + x + x^2 + \dots x^n + \dots = \sum_{n=0}^{\infty} x^n = \frac{1}{1-x}$$

b. The generating function for the sequence $1, 2, 3, 4, \dots$ is given by

$$G(x) = 1 + 2x + 3x^2 + \dots = \sum_{n=0}^{\infty} (n+1)x^n = \frac{1}{(1-x)^2}$$

Problem 1.3.1. Use the method of generating function to solve the recurrence relation

$$a_{n+1} - 8a_n + 16a_{n-1} = 4^n \quad n \geq 1, a_0 = 1, a_1 = 8.$$

Solution: Let the generating function of $\{a_n\}$ be

$$G(x) = a_0 + a_1x + a_2x^2 + \cdots = \sum_{n=0}^{\infty} a_nx^n.$$

Multiplying both sides of the given Recurrence Relation by x^n and summing up, we have

$$\begin{aligned} & \sum_{n=1}^{\infty} a_{n+1}x^n - 8 \sum_{n=1}^{\infty} a_nx^n + 16 \sum_{n=1}^{\infty} a_{n-1}x^n = \sum_{n=1}^{\infty} 4^n x^n \\ \Rightarrow & (a_2x + a_3x^2 + \cdots + a_{n+1}x^n + \cdots) - 8(a_1x + a_2x^2 + \cdots + a_nx^n + \cdots) \\ & + 16(a_0x + a_1x^2 + \cdots + a_{n-1}x^n + \cdots) = \frac{4x}{1-4x} \\ \Rightarrow & \frac{1}{x}(a_2x^2 + a_3x^3 + \cdots + a_{n+1}x^n + \cdots) - 8(a_1x + a_2x^2 + \cdots + a_nx^n + \cdots) \\ & + 16x(a_0 + a_1x + \cdots + a_{n-1}x^n + \cdots) = \frac{4x}{1-4x} \\ \Rightarrow & \frac{1}{x}(G(x) - a_0 - a_1x) - 8(G(x) - a_0) + 16xG(x) = \frac{4x}{1-4x} \\ \Rightarrow & (G(x) - a_0 - a_1x) - 8x(G(x) - a_0) + 16x^2G(x) = \frac{4x^2}{1-4x} \\ \Rightarrow & (1 - 8x + 16x^2)G(x) = \frac{4x^2}{1-4x} + a_0 + a_1x - 8a_0x \\ \Rightarrow & (1 - 8x + 16x^2)G(x) = \frac{4x^2}{1-4x} + 1 \\ \Rightarrow & G(x) = \frac{4x^2}{(1-4x)^3} + \frac{1}{(1-4x)^2} \\ \Rightarrow & G(x) = \frac{1}{4} \frac{1}{(1-4x)^3} + \frac{1}{2} \frac{1}{(1-4x)^2} + \frac{1}{4} \frac{1}{(1-4x)} \\ \Rightarrow & a_0 + a_1x + \cdots + a_nx^n + \cdots = \frac{1}{4}(1.2 + 2.3.(4x) + \cdots + (n+1)(n+2)(4x)^n \dots) \\ & + \frac{1}{2}(1 + 2.4x + 3.(4x)^2 + \cdots + (n+1)(4x)^n \dots) \\ & + \frac{1}{4}(1 + 4x + (4x)^2 + \cdots + (4x)^n \dots). \end{aligned}$$

Then

$$\begin{aligned} a_n &= \text{coefficient of } x^n \text{ in } G(x) \\ &= \frac{1}{4}(n+1)(n+2)4^n + \frac{1}{2}(n+1).4^n + \frac{1}{4}4^n \\ &= (n+1)(n+2)4^{n-1} + 2(n+1).4^{n-1} + 4^{n-1} \\ &= (n^2 + 5n + 5)4^{n-1}. \end{aligned}$$

Problem 1.3.2. Find the generating function of the recurrence relation

$$S(n) - 2S(n-1) - 3S(n-2) = 0, \quad n \geq 2, \quad \text{with } S(0) = 3, S(1) = 1,$$

and hence find its solution.

Solution: Let the generating function of $\{a_n\}$ be

$$G(x) = S(0) + S(1)x + S(2)x^2 + \cdots = \sum_{n=0}^{\infty} S(n)x^n.$$

Multiplying both sides of the given Recurrence Relation by x^n and summing up, we have

$$\begin{aligned}
& \sum_{n=2}^{\infty} S(n)x^n - 2 \sum_{n=2}^{\infty} S(n-1)x^n - 3 \sum_{n=2}^{\infty} S(n-2)x^n = 0 \\
\Rightarrow & (S(2)x^2 + S(3)x^3 + \dots + S(n)x^n + \dots) - 2(S(1)x^2 + S(2)x^3 + \dots + S(n-1)x^n + \dots) \\
& \quad - 3(S(0)x^2 + S(1)x^3 + \dots + S(n-2)x^n + \dots) = 0 \\
\Rightarrow & (G(x) - S(0) - S(1)x) - 2x(S(1)x + S(2)x^2 + \dots + S(n)x^n + \dots) \\
& \quad - 3x^2(S(0) + S(1)x + S(2)x^2 + \dots + S(n)x^n + \dots) = 0 \\
\Rightarrow & (G(x) - 3 - x) - 2x(G(x) - 3) - 3x^2G(x) = 0 \\
\Rightarrow & (1 - 2x - 3x^2)G(x) = 3 + x + 6x \\
\Rightarrow & (1 - 2x - 3x^2)G(x) = 3 + 7x \\
\Rightarrow & G(x) = \frac{3 + 7x}{(1 - 2x - 3x^2)} \\
\Rightarrow & G(x) = \frac{4}{(1 - 3x)} - \frac{1}{(1 + x)} \\
\Rightarrow & a_0 + a_1x + \dots + a_nx^n + \dots = 4(1 + 3x + (3x)^2 + \dots + (3x)^n \dots) \\
& \quad - (1 - x + x^2 - x^3 + \dots + (-1)^n x^n \dots).
\end{aligned}$$

Then

$$\begin{aligned}
a_n &= \text{coefficient of } x^n \text{ in } G(x) \\
&= 4 \cdot 3^n + (-1)^{n+1}.
\end{aligned}$$

Problem 1.3.3. Solve the relation $a_n - 3a_{n-1} = n$ where $n \geq 1$ and $a_0 = 1$ using the method of generating functions.

Solution: Let the generating function of $\{a_n\}$ be

$$G(x) = a_0 + a_1x + a_2x^2 + \dots = \sum_{n=0}^{\infty} a_nx^n.$$

Multiplying both sides of the given Recurrence Relation by x^n and summing up, we have

$$\begin{aligned}
& \sum_{n=1}^{\infty} a_nx^n - 3 \sum_{n=1}^{\infty} a_{n-1}x^n = \sum_{n=1}^{\infty} nx^n \\
\Rightarrow & (a_1x + a_2x^2 + \dots + a_nx^n + \dots) - 3(a_0x + a_1x^2 + \dots + a_{n-1}x^n + \dots) = x + 2x^2 + 3x^3 + \dots + nx^n + \dots \\
\Rightarrow & (G(x) - a_0) - 3x(a_0 + a_1x + a_2x^2 + \dots + a_nx^n) = \frac{x}{(1-x)^2} \\
\Rightarrow & (G(x) - 1) - 3xG(x) = \frac{x}{(1-x)^2} \\
\Rightarrow & (1 - 3x)G(x) = \frac{x}{(1-x)^2} + 1 \\
\Rightarrow & G(x) = \frac{x}{(1-x)^2(1-3x)} + \frac{1}{1-3x} \\
\Rightarrow & G(x) = \frac{7}{4} \frac{1}{(1-3x)} - \frac{1}{4} \frac{1}{(1-x)} - \frac{1}{2} \frac{1}{(1-x)^2} \\
\Rightarrow & a_0 + a_1x + \dots + a_nx^n + \dots = \frac{7}{4}(1 + 3x + (3x)^2 + \dots + (3x)^n + \dots) - \\
& \quad \frac{1}{4}(1 + x + \dots + x^n + \dots) - \frac{1}{2}(1 + 2x + 3x^2 + \dots + (n+1)x^n + \dots)
\end{aligned}$$

Then

$$\begin{aligned}a_n &= \text{coefficient of } x^n \text{ in } G(x) \\&= \frac{7}{4}3^n - \frac{1}{2}(n+1) - \frac{1}{4} \\&= \frac{7}{4}3^n - \frac{1}{2}n - \frac{3}{4}.\end{aligned}$$

1.3 EXERCISES

1. Solve the recurrence relation using generating functions

$$a_{n+2} + 3a_{n+1} + 2a_n = 3^n, \quad n \geq 0, \quad a_0 = 0, \quad a_1 = 1$$

2. Use the method of generating functions to solve the recurrence relation

$$a_n = 4a_{n-1} - 4a_{n-2} + 4^n, \quad n \geq 2, \quad a_0 = 2, \quad a_1 = 8$$

3. Use generating function to solve the recurrence relation

$$a_k = 5a_{k-1} - 6a_{k-2}$$

with initial conditions $a_0 = 6$ and $a_1 = 30$.

4. Using generating function, solve the recurrence relation

$$a_n - a_{n-1} - 6a_{n-2} = 0$$

given $a_0 = 2$ and $a_1 = 1$.

5. Solve the following recurrence relation using generating function

$$a_{n+2} - 8a_{n+1} + 15a_n = 0, \quad \text{given that } a_0 = 2, \quad a_1 = 8.$$

REFERENCES

- [1] T. Veerarajan. Discrete Mathematics. McGraw Hill Education (India) Private Limited, Chennai, 2019.